

Federica Domecq Lacroze

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EDUCATION

Wake Forest University

M.S. in Statistics, Teaching Assistantship

Winston-Salem, NC

Aug. 2026 – May 2028

Pomona College

B.A. Mathematics & Computer Science | Magna Cum Laude, GPA: 3.978/4.00

Claremont, CA

Aug. 2022 – May 2026

Phi Beta Kappa & Sigma Xi Honour Societies

Honours: Bruce Jay Levy Senior Prize in Mathematics; Distinction in Mathematics Senior Exercise; Multi-semester Pomona College Scholar; Bernard Charnwut Chan Leadership & Service Award; Sheri Sangji Memorial Award

Relevant coursework:

- **CS:** Machine Learning; Data Visualisation; Advanced Algorithms; Computer Systems; Data Structures & Advanced Programming; Languages & Theory; Discrete Math & Functional Programming
 - **Math:** Computational Statistics; Methods in Biostatistics; Statistical Theory; Time Series; Deterministic Operations Research; Real Analysis I & II; Probability; Combinatorial Mathematics; Linear Algebra
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EXPERIENCE

Pomona College

Claremont, CA

Kenneth Cooke Research Fellow – “On Par Golf: Leveraging GPR for Perfect Golf”

May 2025 – Aug. 2025

- **National USRESP Winner (1st Prize) & eCOTS 2026 Accepted:** Research selected as national winner; currently developing manuscript for peer-reviewed submission with faculty co-author.
- **Spatial Pipeline:** Digitised a real golf course (satellite imagery via OpenStreetMap, QGIS, Shapely) into a 280-point strategy grid; engineered synthetic 3D green topography with heteroscedastic putting simulation using slope- and tier-based difficulty multipliers.
- **GPR & Multi-Fidelity Fusion:** Built GPyTorch Gaussian Process models to learn continuous Expected-Strokes-to-Hole-Out risk surfaces; implemented a Kennedy–O’Hagan multi-fidelity GP fusing simulated and real location-outcome data to correct simulation bias.
- **Monte Carlo Decision Engine:** Ran Monte Carlo simulations ($N=300$ /decision) with bivariate Gaussian shot dispersion and rotation-matrix aimpoint transforms across two loss functions (expected-strokes minimisation vs. birdie-probability maximisation); validated stability via 100-seed convergence analysis ($N \geq 180$ for 80% stability), independently recovering the literature finding that distance dominates dispersion in expected score.

Senior Thesis (Distinction) in Mathematics

Jan. 2024 – May 2026

- Earned Distinction for “The Math of Moral Compromise, Beyond Arrow’s Impossibility” (Advisor: Prof. Vin de Silva); delivered two independent proofs of Arrow’s Impossibility Theorem (a combinatorial decisive-coalition argument and a topological ultrafilter-based argument), and grounded the theory in an empirical analysis of electoral systems and democratic-freedom indices across 170+ countries.

Statistics/Data Science & Probability Teaching Assistant

Jan. 2024 – May 2026

- Teacher Assistant for Probability (3×), Introductory Statistics (2×), Data Science (1×) over 5 semesters; led sessions on hypothesis testing, estimation, regression, ANOVA, and simulation.
- Mentored students in R/Tidyverse (dplyr, ggplot2, purrr) and data-driven problem-solving; facilitated group learning and personalised code-review.

Summer Research, Professor Assistantship

May 2024 – Aug. 2024

- Applied sigmoidal & impulse models (Sicegar, ImpulseDE2, DESeq2) to gene expression data; parallelised simulations with the **future** package on HPC; co-authored conference talk and academic manuscript.

Resident Advisor

Dec. 2024 – May 2025

- Supervised 100+ students in a 400-resident zone; planned 5+ community programs per semester; resolved conflicts, responded to crises, and liaised with campus staff on student welfare.

DoctorSIM

Madrid, Spain

Marketing & Administrative Intern

Dec. 2021 – Jul. 2022

- Built SEO database of 50+ clients; managed invoicing, DHL shipments, and financial record-keeping.
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ACTIVITIES & LEADERSHIP

Women’s Varsity Golf – 2-Year Team Captain, Pomona-Pitzer

2022 – 2026

- **3× NCAA National Appearances;** 2 Individual National Titles; 2× Team Conference Champions; 3× All-Conference, 2× All-Region; WGCA All-American Scholar 2023, 2024, 2025, 2026.

Co-President, AWM Claremont Chapter

2023 – 2025

Math Department Liaison

2025 – 2026

SKILLS & LANGUAGES

Programming: Python (GPyTorch, Pandas, Shapely); R (Tidyverse); SQL; JavaScript (D3); Haskell; $\text{\LaTeX}$; HTML/CSS

Tools: Git; QGIS; Excel; HPC & Parallel Processing; LLM-assisted coding & documentation

Languages: Fluent English, Fluent Spanish, Intermediate French, Intermediate Portuguese